

Factsheet: List of integrals

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Summary

A list of common (and some uncommon) integrals of functions.

Throughout, a, k are real numbers and C is the constant of integration.

Antiderivatives of polynomial, rational, exponential, logarithmic functions

function	antiderivative w.r.t x	notes
a	$ax + C$	
ax^n	$\frac{ax^{n+1}}{n+1} + C$	$n \in \mathbb{R}, n \neq -1$
ax^{-1}	$a \ln x + C$	
$\frac{a}{bx+c}$	$\frac{a}{b} \ln bx+c + C$	$b, c \in \mathbb{R}$
$\frac{a}{(bx+c)^n}$	$\frac{a(bx+c)^{1-n}}{b(1-n)} + C$	$b, c \in \mathbb{R}$
ae^{kx}	$\frac{a}{k}e^{kx} + C$	
$a \ln(kx)$	$ax \ln kx - ax + C$	

Antiderivatives of trigonometric functions

function	antiderivative w.r.t x
$a \sin(kx)$	$-\frac{a}{k} \cos(kx) + C$
$a \cos(kx)$	$\frac{a}{k} \sin(kx) + C$
$a \tan(kx)$	$\frac{a}{k} \ln \sec(kx) + C$
$a \cot(kx)$	$\frac{a}{k} \ln \sin(kx) + C$
$a \sec(kx)$	$\frac{a}{k} \ln \tan(kx) + \sec(kx) + C$
$a \csc(kx)$	$\frac{a}{k} (\ln \sin(\frac{kx}{2}) - \ln \cos(\frac{kx}{2})) + C$

Antiderivatives of some hyperbolic functions

function	antiderivative w.r.t x
$a \sinh(kx)$	$\frac{a}{k} \cosh(kx) + C$
$a \cosh(kx)$	$\frac{a}{k} \sinh(kx) + C$
$a \tanh(kx)$	$\frac{a}{k} \ln \cosh(kx) + C$
$a \coth(kx)$	$\frac{a}{k} \ln \sinh(kx) + C$

Standard forms that integrate to inverse trigonometric/hyperbolic functions

function	antiderivative w.r.t x
$\frac{a}{\sqrt{1 - k^2 x^2}}$	$\frac{a}{k} \sin^{-1}(kx) + C$

function	antiderivative w.r.t x
$-\frac{a}{\sqrt{1-k^2x^2}}$	$\frac{a}{k} \cos^{-1}(kx) + C$
$\frac{a}{1+k^2x^2}$	$\frac{a}{k} \tan^{-1}(kx) + C$
$\frac{a}{\sqrt{1+k^2x^2}}$	$\frac{a}{k} \sinh^{-1}(kx) + C$
$\frac{a}{\sqrt{k^2x^2-1}}$	$\frac{a}{k} \cosh^{-1}(kx) + C$
$\frac{a}{\sqrt{1-k^2x^2}}$	$\frac{a}{k} \tanh^{-1}(kx) + C$

Further reading

For more about where these came from, please see [Guide: Introduction to integration](#) and [Proof: Antiderivatives of other common functions].

Version history

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